

Even & Odd Functions

Even Functions:

A function is even if replacing x with $-x$ doesn't change the output.

$$f(-x) = f(x)$$

These functions are symmetrical about y -axis.

Examples: x^2 , $\cos(x)$, $|x|$.

For Even Functions,

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

Odd Functions:

A function is odd if replacing x with $-x$ changes the output.

$$f(-x) = -f(x)$$

These functions are symmetrical about origin.

Examples: x , x^3 , $\sin(x)$.

For Odd Functions,

$$\int_{-a}^a f(x) dx = 0$$

Half-Range Fourier Series (Sine & Cosine) or Fourier Series for Even & Odd Functions

1. If $f(x)$ is an even function,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) \quad [0 < x < L]$$

$$a_0 = \frac{2}{L} \int_0^L f(x) dx \quad \& \quad a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \quad [0 < x < \pi]$$

Cosine Series

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx \quad \& \quad a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos n\pi dx$$

2. If $f(x)$ is an odd function,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) \quad [0 < x < L]$$

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx \quad [0 < x < \pi]$$

Sine Series

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$$

Q.1) Find a series of cosine of multiple of x which will represent $f(x)$ in $(0, \pi)$ where,

$$f(x) = \begin{cases} 0 & 0 < x < \pi/2 \\ \pi/2 & \pi/2 < x < \pi \end{cases}$$

Deduce that: $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \infty = \pi/4$

Solⁿ:

~~For~~ fourier cosine series is,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

Now,

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$= \frac{2}{\pi} \int_0^{\pi/2} f(x) dx + \frac{2}{\pi} \int_{\pi/2}^{\pi} f(x) dx$$

$$= 0 + \frac{2}{\pi} \int_{\pi/2}^{\pi} \frac{\pi}{2} dx$$

$$= \left[x \right]_{\pi/2}^{\pi}$$

$$= \frac{\pi}{2}$$

Also,

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi/2} 0 \cdot \cos nx \, dx + \frac{2}{\pi} \int_{\pi/2}^{\pi} \frac{\pi}{2} \cos nx \, dx$$

$$= 0 + \left[\frac{\sin nx}{n} \right]_{\pi/2}^{\pi}$$

$$= \frac{\sin n\pi}{n} - \frac{\sin n\pi/2}{n}$$

$$= -\frac{1}{n} \sin\left(\frac{n\pi}{2}\right)$$

$$\therefore f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \left(-\frac{1}{n} \sin\left(\frac{n\pi}{2}\right) \right) \cos nx$$

$$\Rightarrow f(x) = \frac{\pi}{4} - \left(\sin \frac{\pi}{2} \cdot \cos x + \frac{1}{2} \sin \pi \cdot \cos 2x + \frac{1}{3} \sin \frac{3\pi}{2} \cdot \cos 3x + \dots \right)$$

$$\Rightarrow f(x) = \frac{\pi}{4} - \left[\cos x - \frac{1}{3} \cos 3x + \frac{1}{5} \cos 5x + \dots \right]$$

$$\Rightarrow f(0) = \pi/4 - (1 - 1/3 + 1/5 - 1/7 + \dots) \quad [x=0 \text{ (say)}]$$

$$\therefore 1 - 1/3 + 1/5 - 1/7 + \dots = \pi/4 \quad \text{proved}$$

Q.2) Express $f(x) = x$ as a sine series in $0 < x < \pi$.

Solⁿ:

The fourier sine series is,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

Now,

$$b_n = \frac{2}{\pi} \int_0^{\pi} \underbrace{x}_{\text{I}} \underbrace{\sin(nx)}_{\text{II}} dx$$

$$= \frac{2}{\pi} \left[\left[x \left(\frac{-\cos nx}{n} \right) \right]_0^{\pi} - \int_0^{\pi} (1) \left(\frac{-\cos nx}{n} \right) dx \right]$$

$$= \frac{2}{\pi} \left\{ \left[\frac{-x \cos nx}{n} \right]_0^{\pi} + \left[\frac{\sin nx}{n^2} \right]_0^{\pi} \right\}$$

$$= \frac{2}{\pi} \left\{ \frac{-\pi \cos n\pi}{n} + 0 + \frac{\sin n\pi}{n^2} - 0 \right\}$$

$$= \frac{-2}{n} \cos n\pi$$

$$\therefore f(x) = \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n} \sin nx$$

$$= \frac{-2(-1)^n}{n}$$

$$\Rightarrow x = 2 \left[\sin x + \frac{1}{2} \sin 2x + \dots \right]$$

$$= \frac{2(-1)^{n+1}}{n}$$

Q.3) Find the fourier series expansion of the function $f(x) = \cos(sx)$, $-\pi \leq x \leq \pi$ where s is a fraction.

Hence, show that $\cot \alpha = \frac{1}{\alpha} + \frac{2\alpha}{\alpha^2 - \pi^2} + \frac{2\alpha}{\alpha^2 - 4\pi^2} + \dots$

Solⁿ:

The fourier cosine series is,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

Now,

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \cos sx dx$$

$$= \frac{2}{\pi} \left[\frac{\sin sx}{s} \right]_0^{\pi}$$

$$= \frac{2}{\pi s} \sin s\pi$$

Also,

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \cos sx \cdot \cos nx \, dx$$

$$= \frac{1}{\pi} \int_0^{\pi} [\cos(sx+nx) + \cos(sx-nx)] \, dx$$

$$[\because 2 \cos A \cos B = \cos(A+B) + \cos(A-B)]$$

$$= \frac{1}{\pi} \left[\frac{\sin(st+n)x}{st+n} + \frac{\sin(s-n)x}{s-n} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{\sin(st+n)\pi}{st+n} + \frac{\sin(s-n)\pi}{s-n} - 0 - 0 \right]$$

$$\therefore f(x) = \frac{1}{\pi s} \cdot \sin s\pi + \sum_{n=1}^{\infty} \frac{1}{\pi} \left[\frac{\sin(st+n)\pi}{st+n} + \frac{\sin(s-n)\pi}{s-n} \right] \cdot (\cos nx)$$

Put $x = \pi$,

$$\Rightarrow \cos(s\pi) = \frac{1}{\pi s} \sin s\pi + \frac{1}{\pi} \sum_{n=1}^{\infty} \left[\frac{\sin(st+n)\pi}{st+n} + \frac{\sin(s-n)\pi}{s-n} \right] (-1)^n$$

$$= \frac{1}{\pi s} \sin s\pi + \frac{1}{\pi} \left[- \left(\frac{\sin(st+1)\pi}{st+1} + \frac{\sin(s-1)\pi}{s-1} \right) + \left(\frac{\sin(st+2)\pi}{st+2} + \frac{\sin(s-2)\pi}{s-2} \right) + \dots \right]$$

$$= \frac{\sin s\pi}{\pi s} + \frac{1}{\pi} \left[-\left(\frac{\sin(s\pi + \pi)}{s+1} + \frac{\sin(s\pi - \pi)}{s-1} \right) + \left(\frac{\sin(s\pi + 2\pi)}{s+2} + \frac{\sin(s\pi - 2\pi)}{s-2} \right) + \dots \right]$$

$$= \frac{\sin s\pi}{\pi s} + \frac{1}{\pi} \left[-\left(-\frac{\sin s\pi}{s+1} - \frac{\sin s\pi}{s-1} \right) + \left(\frac{\sin s\pi}{s+2} + \frac{\sin s\pi}{s-2} \right) + \dots \right]$$

$$\left[\begin{array}{l} \sin(\pi + \theta) = -\sin \theta, \quad \sin(\pi - \theta) = \sin \theta \\ \sin(2\pi + \theta) = \sin \theta, \quad \sin(2\pi - \theta) = -\sin \theta \end{array} \right]$$

$$= \frac{\sin s\pi}{\pi s} + \frac{\sin s\pi}{\pi} \left[\left(\frac{1}{s+1} + \frac{1}{s-1} \right) + \left(\frac{1}{s+2} + \frac{1}{s-2} \right) + \dots \right]$$

$$= \frac{\sin s\pi}{\pi s} + \frac{\sin s\pi}{\pi} \left[\frac{s-1+s+1}{s^2-1^2} + \frac{s-2+s+2}{s^2-2^2} + \dots \right]$$

$$= \frac{\sin s\pi}{\pi s} + \frac{\sin s\pi}{\pi^2} \left[\frac{2s}{s^2-1^2} + \frac{2s}{s^2-2^2} + \dots \right]$$

$$= \frac{\sin s\pi}{\pi s} + \sin s\pi \left[\frac{2s\pi}{s^2\pi^2-\pi^2} + \frac{2s\pi}{s^2\pi^2-2^2\pi^2} + \dots \right]$$

$$= \sin s\pi \left[\frac{1}{\pi s} + \frac{2s\pi}{s^2\pi^2-\pi^2} + \frac{2s\pi}{s^2\pi^2-2^2\pi^2} + \dots \right]$$

Putting, $S\pi = \alpha$

$$\cos \alpha = \sin \alpha \left[\frac{1}{\alpha} + \frac{2\alpha}{\alpha^2 - \pi^2} + \frac{2\alpha}{\alpha^2 - 4\pi^2} + \dots \right]$$

$$\Rightarrow \cot \alpha = \left[\frac{1}{\alpha} + \frac{2\alpha}{\alpha^2 - \pi^2} + \frac{2\alpha}{\alpha^2 - 4\pi^2} + \dots \right]$$

proved.

$$\left[\therefore \frac{\cos \theta}{\sin \theta} = \cot \theta \right]$$